



Large Scale Hybrid Computing

2026

André Pereira

Course Syllabus



Semana	Data	Teórica		Prática Laboratorial	
	02/02 a 07/02				
1	09/02 a 14/02	Presentation an evaluation methodology. Overview of current supercomputers and their manycore chips.	2h		1h
2	16/02 a 21/02	Advanced MPI - one sided communication.	1h30	Lab 1: Advanced MPI	1h30
3	23/02 a 28/02	Advanced MPI - hybrid computing.	1h30	Lab 2: MPI and multithreading	1h30
4	02/03 a 07/03	Profiling MPI applications at scale.	1h	Lab 3: Profiling hybrid MPI code at scale	2h
5	09/03 a 14/03	The PCAM methodology.	1h	Lab 4: Domain and task partitioning with MPI	2h
6	16/03 a 21/03	First work assignment defense.	3h	-	0h
7	23/03 a 28/03	Graph theory. Directed acyclic graphs.	1h30	Lab 5: Parallel graph algorithms	1h30
	30/03 a 04/04				
8	06/04 a 11/04	Scheduling of regular and irregular workloads.	1h	Lab 6: Scheduling in MPI applications	2h
9	13/04 a 18/04	Support to the second work assignment.	3h	-	0h
10	20/04 a 25/04	Energy efficient computing at scale.	1h	Lab 7: Energy profile in multi-node usage	2h
11	27/04 a 02/05	Buffer	1h		2h
12	04/05 a 09/05	Disaggregated memory in High Performance Computing.	2h	Lab 8: Assessing simulated memory pools	1h
	11/05 a 16/05				
	18/05 a 23/05	Exam	3h	-	0h
	25/05 a 30/05	Entrega do 2º trabalho dia 4 de Janeiro		Defesa do 2º trabalho dia 9 de Janeiro	



- Work assignment 1 (15%)
 - Paper analysis and presentation
 - Subject of WA 2
- Work assignment 2 (55%)
 - Analysis, profiling, and characterisation of the performance and limitations of an exascale flagship code
 - No code development, but requires testing different configurations
- Written test (30%)



- Work assignment 1 (15%)
 - Presentation and defence on March 19th
- Work assignment 2 (55%)
 - Submission on May 24th at 23h59
 - Presentation and defence on week 15 - TBD
- Written test (30%)
 - May 28th